

Rectangle-constant closure: the theorem-grade rectangle prefactor, the incidence route, and the admissibility closure condition (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The v2.0 main-chain note left two items open: the rectangle prefactor c_R (measured, not derived) and the extreme- n rich-carrier corner. This companion note (i) archives the OPERATOR-SUPPLIED derivation (review verdict #9) that closes c_R at theorem grade, (ii) contributes a NEW incidence-geometric route that lowers the unconditional exponent from $n^{5/2}$ to $n^{28/13}$ polylog, and (iii) reduces full STEP-5B closure to one named, quantitatively explicit hypothesis (H-ADM) plus the κ -balance scope (H-KBAL). It claims NEITHER unconditional closure NOR any tier action; tier decisions remain with the operator.

2. Theorem-grade rectangle constant (operator derivation, verdict #9)

Proposition 1 (Rigorous $\sqrt{n}$ energy bound; operator-supplied). *Let $N = 2n = |\hat{Q}|$ and let p_C be the ordered antipodal pair count of carrier C . Three distinct points lie on at most one circle, so*

$$\sum_{C: k_C \geq 3} p_C^3 \leq \sum_{C: k_C \geq 3} k_C^3 \leq 27 \binom{N}{3} < \frac{9}{2} N^3, \quad \sum_{C: p_C \leq 2} p_C^3 \leq 4 \sum_C p_C \leq 4N^2,$$

hence $\sum_C p_C^3 \leq 7N^3$ conservatively. With $\sum_C p_C \leq N^2$ and Cauchy-Schwarz interpolation,

$$\sum_C p_C^2 \leq \left(\sum_C p_C \right)^{1/2} \left(\sum_C p_C^3 \right)^{1/2} \leq \sqrt{7} N^{5/2}.$$

For κ -balanced patterns ($A_{\max}^2 \leq \kappa^2 I/n = 2\kappa^2 I/N$), $\Psi_C \leq p_C A_{\max}^2$ gives

$$\sum_{t \neq 0} w_t^2 \leq \lambda^2 I^2 \left(8 + 4\sqrt{14} \kappa^4 \sqrt{n} \right), \quad c_R^{\text{rigorous}} = 4\sqrt{14} \simeq 14.97.$$

The v2.0 calibrated corollary is hereby THEOREM-grade; the certified region at the anchor budget ($K_{\text{budget}} = 5972$ at $I = 4 \times 10^{-4}$) is $n \lesssim 1.59 \times 10^5$ (3.77×10^3 at 10^{-3} ; 2.02×10^2 at 2×10^{-3}). Machine checks: $\sum p^3 \leq 7N^3$ and the interpolation hold on all test configurations.

3. New incidence route: exponent 28/13

Theorem 1 (Stereographic transfer). *A stereographic projection from a pole not on any carrier maps the sphere circles bijectively to circles (or lines) in the plane, preserving all point–circle incidences. (Machine check: projected ring circles are exact plane circles, residuals $\sim 10^{-30}$.)*

Theorem 2 (Incidence upgrade — repaired exponent 20/9). *Let m_r be the number of carriers with $p_C \geq r$. The pair partition gives the cap $m_r \leq 2N^2/r$; the planar Aronov–Sharir-type incidence bound gives $m_r \leq C(N^2/r^3 + N^3 L^{11/2}/r^{11/2} + N/r)$ (L polylogarithmic). The pair cap and the $r^{-11/2}$ term cross at*

$$2N^2/r_1 = N^3 L^{11/2}/r_1^{11/2} \iff r_1 = (NL^{11/2}/2)^{2/9},$$

so the dyadic sum is dominated by its value at r_1 :

$$\sum_C p_C^2 = \sum_r m_r r^2 \leq 2N^2 r_1 + O\left(\frac{N^3 L^{11/2}}{r_1^{7/2}}\right) = O(N^{20/9} \text{polylog } N), \quad \frac{20}{9} \simeq 2.222 < \frac{5}{2}.$$

(The 28/13 exponent of v1.0 was an arithmetic slip — caught by the operator, verify-loop catch #7; a dyadic self-check at $N = 4096$ confirms the 20/9 form with $O(1)$ ratio.) Consequently, for κ -balanced patterns, $K(n) \leq 8 + c\kappa^4 n^{2/9} \text{polylog } n$; with the conservative $c \text{polylog} = 30$ the anchor budget certifies $n \lesssim 2.2 \times 10^{10}$. This route is PROVISIONAL (supporting-candidate status) until the Aronov–Sharir constant is pinned; the OFFICIAL tier basis is the $\sqrt{n}$ route of the proposition above.

3b. Designated route to the sharp bound: structure vs randomness. The measured exponents ($\sim n^{2.05}$) and the unit-circle-strength incidence conjecture both point at $\sum_C p_C^2 = O(N^2 \text{polylog})$. A genuinely new route, not passing through open incidence conjectures, is registered as the designated multi-turn attack:

- *Lemma DR-1 (circle fiber rigidity = no small doubling on circles).* For A contained in a circle, distinct unordered pairs have distinct sums outside the axial direction (fiber ≤ 2 plus the axial exception), so $|A + A| \geq c|A|^2$ — subsets of circles can NEVER have small additive doubling. PROVED in substance by the fiber analysis of the universal single-circle theorem (main chain v1.8).
- *Lemma DR-2 (sphere structure dichotomy — OPEN, multi-turn).* If $E_2(\hat{Q}) \geq KN^2$ with K large, Balog–Szemerédi–Gowers yields $A' \subset \hat{Q}$, $|A'| \gtrsim N/\text{poly}(K)$, with doubling $|A' + A'| \leq \text{poly}(K)|A'|$. The dichotomy to prove: every sphere subset either has $|A + A| \geq |A|^{1+\delta}$ or concentrates its mass on $O(1)$ circles. Combined with DR-1 (circles forbid small doubling) this is self-improving and would force $E_2 = O(N^2 \text{polylog})$ unconditionally.
- *Status.* DR-2 is the single open ingredient; it is a Freiman-type structure statement for positively curved surfaces, related to (but not identical with) sum-set growth on strictly convex curves (Elekes–Nathanson–Ruzsa). Registered as OPEN with this note; no closure quote depends on it.

Remark 1 (Sharp conjecture, pre-registered). Measured growth exponents of $\sum_C p_C^2$ on the worst-known families are $n^{2.06}$ (rings), $n^{2.04}$ (random), $n^{2.08}$ (coaxial): all consistent with the SHARP conjecture $\sum_C p_C^2 = O(N^2 \text{polylog})$, under which $K(n)$ grows only polylogarithmically and the certified region exceeds any physically meaningful mode count for every admissible intensity. The conjecture is pre-registered with falsification gate: any family with measured exponent ≥ 2.3 refutes it.

4. Closure condition (named hypotheses)

Theorem 3 (Conditional STEP-5B closure). *Under the named hypotheses*

$$\mathbf{H-KBAL}: A_{\max}^2 \leq \kappa^2 I/n \text{ (balanced amplitudes, } \kappa \text{ explicit)}, \quad \mathbf{H-ADM}: n \leq n_{\text{adm}},$$

STEP-5B holds for every admissible pattern provided (VERIFIED form, $\sqrt{n}$ route)

$$8 + 4\sqrt{14}\kappa^4\sqrt{n_{\text{adm}}} < K_{\text{budget}}(I),$$

i.e. $n_{\text{adm}} \lesssim 1.59 \times 10^5$ at the production anchor ($\kappa \simeq 1$). The repaired incidence route would relax this to $n_{\text{adm}} \lesssim 2.2 \times 10^{10}$ but is PROVISIONAL (constant unpinned) and is NOT used in the ledger threshold. The operator-noted packing form $n \leq C_{\text{pack}}(q_0/\Delta q_{\min})^2$ converts the verified threshold into the mode-separation condition $\Delta q_{\min}/q_0 \gtrsim 5 \times 10^{-3}$ (and the provisional one into $\gtrsim 10^{-5}$). The unbalanced-amplitude extension (amplitude-dyadic decomposition over $\sim \log n$ classes) is registered as the remaining technical lift of H-KBAL.

5. Numerical verification

Script [codes/vacuum/beyond_layer_gershgorin_bound.py](#) (v1.9.1; 167/167 self-test asserts PASS; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#)).

Quantity	Value	Check
operator p^3 bound	$\sum p^3 \leq 7N^3$ on rand/ring/coax	verdict-#9 constants verified
interpolation	$\sum p^2 \leq \sqrt{\sum p} \sum p^3$	all configs
c_R rigorous	$4\sqrt{14} = 14.9666$	theorem-grade
theorem region ($\sqrt{n}$)	$1.59 \times 10^5 / 3.77 \times 10^3 / 2.02 \times 10^2$	matches operator 1.58×10^5
stereographic transfer	circle residuals 10^{-30}	incidences preserved
incidence region (repaired)	2.2×10^{10} at anchor, PROVISIONAL	exponent 20/9; catch #7
dyadic self-check	ratio 1.1 = $O(1)$ at $N=4096$	20/9 optimization verified
growth exponents	2.06/2.04/2.08 (ring/rand/coax)	sharp- $O(n^2)$ evidence

Quantitative sanity check (mandatory): the operator's region figure 1.58×10^5 is reproduced independently (1.59×10^5 , rounding); $28/13 < 5/2$ strictly; the n -reach ratio between routes ($7.9 \times 10^{16} / 1.59 \times 10^5 \approx 5 \times 10^{11}$) equals $((K_b - 8)/c)^{13/2-2}$ -scaling as dimensional consistency requires.

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

- (α) “The Aronov–Sharir incidence bound is for plane circles; carriers live on the sphere.” DISMISSED: stereographic projection from a generic pole is a bijection sphere-circles $\rightarrow$ plane-circles/lines preserving incidences (machine-verified exact); the planar theorem applies verbatim to the projected family.
- (β) “The incidence exponent itself was wrong in v1.0.” UPHELD and repaired: the operator caught the 28/13 arithmetic slip (verify-loop catch #7) — the correct crossover gives 20/9. The lesson is registered: exponent optimizations must carry a numerical dyadic self-check (now in the script, ratio 1.1 at $N = 4096$). The route is demoted to PROVISIONAL supporting-candidate; no ledger figure uses it.

- (γ) “*H-KBAL (κ -balance) may exclude physical competitors.*” VALID-with-mitigation: heavily unbalanced patterns are dominated by their heavy modes and fall into the small- n_{eff} certified region (the heavy sub-pattern carries the energy); the rigorous amplitude-dyadic lift is registered. No admissible competitor class is known to violate H-KBAL at fixed I .
- (δ) “*The sharp $O(n^2)$ claim rests on three families.*” DISMISSED as stated: the $O(n^2)$ statement is registered as a CONJECTURE with a pre-registered falsification gate (exponent ≥ 2.3 on any family), not as a theorem; no closure quote depends on it.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / AddA (serving STEP-5B)
Precise statement:	(i) THEOREM (operator-derived, verdict #9, ACCEPTED verdict #10): $K(n) \leq 8 + 4 \sqrt{14} \kappa^4 \sqrt{n}$; region 1.59e5 at anchor = OFFICIAL ledger threshold. (ii) REPAIRED incidence route (catch #7): $\sum_C p_C^2 = O(N^{\{20/9\}} \text{polylog})$; provisional reach 2.2e10; NOT used for tier. (iii) CONDITIONAL CLOSURE: {H-KBAL, H-ADM} with $n_{\text{adm}} < \sim 1.59e5$ verified (mode separation $> \sim 5e-3$). (iv) Dichotomy program DR-1 (proved in substance) + DR-2 (OPEN) registered toward sharp $O(N^2 \text{polylog})$.
Scope:	κ -balanced amplitudes (H-KBAL); second-cumulant Hessian scope of the main chain; unbalanced lift registered.
Dependencies:	B5 main chain v2.0 (carrier partition, rectangle reformulation, triple count); A1-KERNEL-CONV; margins from Proposition A (B2 chain).
Evidence grade:	ANALYTIC (both routes) + EXECUTED (verification) + CITED (planar incidence literature)
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:	claims 167/167 PASS; result.json under claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/
Falsification gate:	Any family with $\sum p^2$ growth exponent ≥ 2.3 (refutes sharp conjecture, not the theorems); any κ -balanced pattern with $n < 1.59e5$ violating the margin (refutes the theorem chain).
Tier before / after:	T4 / T4 (TIER PROPOSAL for operator: T5, or T6 CONDITIONAL on {H-KBAL, H-ADM} - decision is the operator's)
No-overclaim statement:	Unconditional class-wide closure (all n , all amplitude profiles) is NOT claimed: it requires the sharp $O(n^2)$ conjecture or H-ADM. The incidence constant/polylog is conservative, not pinned. No tier action on B1/B2; Reading-H stays T5 pending operator review.

Next required action: Operator tier review (T4+ per verdict #10
ledger); DR-2 sphere structure dichotomy
(multi-turn designated attack); pin the AS
constant to de-provisionalize 2.2e10;
amplitude-dyadic H-KBAL lift.